Paper ID : 250340

Roll No: | | | | | | | | | | | | | |

MCA
(SEM IV) THEORY EXAMINATION 2024-25
BLOCKCHAIN ARCHITECTURE

TIME: 3 HRS **M.MARKS: 100**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 10 = 20**

Q No.	Question	CO	Level
a.	Define blockchain.	1	K1
b.	What is a digital signature?	1	K1
c.	List any two design primitives in blockchain.	1	K2
d.	Define consensus in the context of blockchain.	2	K1
e.	Mention any two consensus mechanisms.	2	K2
f.	What is PoW in blockchain?	2	K2
g.	What is chaincode in Hyperledger Fabric?	4	K1
h.	List two applications of blockchain in supply chain.	5	K4
i.	What is Hyperledger Composer?	4	K2
j.	Mention two features of blockchain for public governance.	5	K4

SECTION B

2. Attempt any *three* of the following: **10 x 3 = 30**

Q No.	Question	CO	Level
a.	Explain the design primitives of blockchain architecture with relevant examples.	1	K2
b.	Compare permissioned and permissionless blockchains with use cases.	2	K4
c.	Describe the scalability issues in blockchain consensus protocols.	2	K4
d.	Describe the architecture and components of Hyperledger Fabric.	4	K1
e.	Explain the role of blockchain in financial software systems such as insurance and KYC.	5	K4

SECTION C

3. Attempt any *one* part of the following: **10 x 1 = 10**

Q No.	Question	CO	Level
a.	Describe the evolution from digital money to distributed ledgers	1	K1
b.	Explain Bitcoin basics and the use of hashchains in blockchain.	1	K2

4. Attempt any *one* part of the following: **10 x 1 = 10**

Q No.	Question	CO	Level
a.	What are the key requirements for consensus protocols? Explain with examples.	2	K4
b.	Describe the Proof of Work mechanism with an example from Bitcoin.	2	K4

QP25EP1_311 | 26-May-2025 9:08:46 AM | 115.245.67.130

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5. Attempt any *one* part of the following: 10 x 1 = 10

Q No.	Question	CO	Level
a.	Illustrate the process of chaincode implementation in Hyperledger Fabric.	4	K1
b.	Discuss the role and structure of the Hyperledger Fabric SDK and Composer tool.	4	K2

6. Attempt any *one* part of the following: 10 x 1 = 10

Q No.	Question	CO	Level
a.	Explain the use of blockchain in capital markets and insurance with real-world scenarios.	5	K5
b.	How can blockchain improve provenance tracking in supply chains?	5	K5

7. Attempt any *one* part of the following: 10 x 1 = 10

Q No.	Question	CO	Level
a.	Discuss how blockchain can be utilized for land record management by the government.	5	K5
b.	Explain privacy and cryptographic concerns in blockchain-based government systems.	5	K5

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